FSDA-DETR: Few-Shot Domain-Adaptive Object Detection Transformer in Remote Sensing Imagery

Binbin Yang, Jianhong Han, Xinghai Hou, *Graduate Student Member, IEEE*, Dehao Zhou, Wenkai Liu, and Fukun Bi

***Abstract*— Few-shot domain-adaptive object detection (FSDAOD) aims to transfer knowledge from a source domain to a target domain with limited labeled data, which faces severe challenges in the field of remote sensing. To address this issue, numerous convolutional neural network (CNN)-based domain adaptation methods employ style transfer and feature alignment to mitigate domain shifts, but limited target domain samples are prone to yielding equivocal optimization and are maladaptive. Furthermore, the sparsity of targets and the complexity of backgrounds in remote sensing imagery contribute to confusing feature alignment. Moreover, detection transformer (DETR)-based detectors have achieved remarkable progress in unsupervised domain adaptation (UDA) but remain unexplored in FSDAOD. To address these challenges, we introduce FSDA-DETR, the first DETR-based strong baseline designed for the FSDAOD of remote sensing imagery. Specifically, we propose a cross-domain style rectification (CSR) module that rectifies the styles of the target domain to align with the source domain by storing and dynamically updating the weighted-fusion source-domain style bases. To further strengthen the detector's cross-domain detection performance, we propose a category-aware feature alignment (CFA) module that performs fine-grained masking on object regions of rectified backbone features corresponding to different categories and utilizes adversarial training for domain-invariant feature extraction. Extensive experiments on three cross-domain benchmarks, comprising six diverse datasets, demonstrate that FSDA-DETR outperforms state-of-the-art methods. For instance, in the optical-to-synthetic aperture radar (SAR) benchmark, FSDA-DETR achieves 74.5% mean average precision (mAP) with only 1% of target-domain training data. The code and datasets are available at https://github.com/wsybb252237/FSDA-DETR**

***Index Terms*— Domain adaptive, few-shot, object detection, remote sensing imagery.**

I. Introduction

OVER the past few decades, the remarkable achievements of deep learning technologies [1], [2], [3], [4] have profoundly advanced object detection in remote sensing imagery. However, the exceptional performance of these models typically hinges on the assumption that the training data (source domain) and testing data (target domain) are independent and identically distributed. In real-world deployments, datasets frequently fail to fully meet the condition of identical distributions, resulting in suboptimal performance of object detectors on target domains with differing distributions. Unsupervised domain adaptation (UDA) [5], [6] offers an effective solution to such challenges. Nonetheless, most UDA algorithms [7] presuppose a small domain shift and the availability of abundant unlabeled target-domain data. In remote sensing, the significant cost associated with acquiring large volumes of target-domain data [8], combined with pronounced domain shifts [9] between distributions, leads to a substantial decline in UDA performance. Consequently, the importance of exploring few-shot domain adaptation (FDA) has become increasingly prominent.

Unlike UDA, few-shot domain-adaptive object detection (FSDAOD) [10], [11], [12] aims to achieve effective domain adaptation of object detectors between the source and target domains with minimal cost. Given the negligible cost of obtaining a small number of finely annotated samples from the target domain, FSDAOD assumes [11] that the availability of sufficient annotated data in the source domain alongside only a few annotated target domain samples (typically eight randomly selected images) that are available for detector adjustment. Following the methodology of Gao et al. [11], the source and target domains share the same category set but exhibit significant domain shifts. In addition, following [13], we further investigate our method under the few-shot setting (i.e., three, five, and ten shots) to verify its generality.

In FSDAOD, most methods based on convolutional neural networks (CNNs), such as [14], [15], style transfer methodologies, and feature alignment strategies, are used to bridge domain shifts. However, under FDA settings, the scarcity of labeled samples and significant domain shifts between the source and target domains are prone to producing equivocal optimization results, resulting in maladaptive problems. Moreover, the significant domain shifts in remote sensing images, characterized by sparse targets and complex backgrounds, often lead to feature alignment results that favor complex background features, thereby yielding suboptimal performance. As a result, directly applying these CNN-based detectors, which are designed for natural

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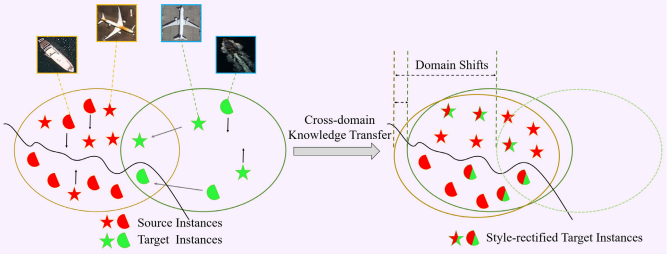


Fig. 1. We address the task of FSDAOD for remote sensing imagery. The overall architecture of the FSDA-DETR. Given the scarcity of target-domain data, a pronounced domain shift is observed between the source and target domains. To address this, we strategically facilitate feature alignment across distinct categories at both the style and feature representation levels, thereby enabling effective knowledge transfer from the source domain to the target domain and ultimately mitigating domain shifts.

scenes, to the remote sensing domain leads to suboptimal performance.

Different from convolutional neural network (CNN)-based detection architectures, detection transformer (DETR)-based detectors [4], [16], [17] due to their end-to-end design to achieve outstanding performance in in-domain scenarios, making them widely adopted in UDA research. The encoder–decoder structure of DETR provides the advantages of self-attention and global context modeling, which we consider advantageous for acquiring domain-invariant features in FSDAOD. Additionally, DETR simplifies the detection process by discarding complex designs such as anchor generation [1] and nonmaximum suppression (NMS) [18], further enhancing its adaptability and robustness in remote sensing object detection tasks. Furthermore, most existing FSDAOD studies focus on image classification [19], with relatively limited exploration in the area of remote sensing object detection [13]. Therefore, exploring the application of DETR-based detectors for FSDAOD in remote sensing images is particularly important.

To address these challenges, we propose a powerful DETR-based baseline, named FSDA-DETR, designed for FSDAOD in remote sensing imagery. Specifically, as shown in Fig. 1, to maximize the utility of few-shot annotated target domain data and alleviate domain shifts driven by visual appearance discrepancies, we introduce an innovative cross-domain style rectification (CSR) module. This module rectifies the target-domain style to align it with the source-domain's style representation space, effectively mitigating the domain shifts of visual styles. Furthermore, to mitigate confusing feature alignment, we design a category-aware feature alignment (CFA) module that performs fine-grained masking on object regions of rectified backbone features corresponding to different categories and utilizes adversarial training to extract domain-invariant features, which further enhances the detector's cross-domain detection performance.

The motivation of the CSR module stems from the ingenious utilization of few-shot target domain samples, alleviating domain shifts through adaptive cross-domain image style transformation. This module is inspired by previous studies [20], [21], which have demonstrated that visual appearance (style) differences between source- and target-domain data are key contributors to the domain shifts in FSDAOD. Unlike their adversarial style training approaches, we ingeniously utilize few-shot annotated target-domain data to project target-domain styles into a style representation space constructed based on source-domain styles. This encourages the model to identify and exploit features relevant to the target domain while persistently updating the stored source-domain style bases. Additionally, the CSR module rectifies the styles of diverse target domains to the source-domain styles, thereby significantly improving the detector's performance in FSDAOD tasks. Specifically, we begin by capturing source-domain style features through commonly used channelwise statistics (mean and variance), which are stored as the basis for style representation [22]. During domain adaptation, the style features of several target-domain images are subsequently projected onto the source-domain feature space. This projection is realized via a weighted combination of stored source-domain style bases, with cross-domain feature statistics estimated through momentum updates across the entire source-domain style. Ultimately, dynamic style rectification of the target domain is enabled through the implementation of the adaptive instance normalization (AdaIN) [23] function.

The CFA module is intended to steer the detector toward focusing on pivotal object regions across distinct categories within backbone feature representations, which effectively addresses the challenge of confusing feature alignment resulting from sparse objects and complex background noise in remote sensing imagery. Unlike the CSR module, the CFA applies fine-grained category-aware masks at specific feature levels, ensuring that the detector extracts domain-invariant instance-level foreground features. Specifically, the availability of annotations in both the source and few-shot target domains enables the CFA module to generate fine-grained, category-aware masks. These masks are subsequently applied to the backbone feature maps from both domains to enhance domain alignment. These category masks are subsequently downsampled to cover the corresponding instance regions accurately, allowing for precise masking operations. This mechanism effectively reduces interference from complex background noise in remote sensing images during feature alignment, ensuring that the model focuses exclusively on the object areas of relevant categories. Finally, the masked feature maps from both the source and target domains are fed into the domain discriminator to achieve cross-domain CFA. This global alignment of backbone features further alleviates domain shifts of cross-domain foreground features.

The main contributions of this article are as follows.

1) We propose FSDA-DETR, a powerful DETR-based baseline for FSDAOD in remote sensing imagery. To the best of authors' knowledge, this is the first attempt to explore DETR-based detectors for FDA in remote sensing images.
2) We design a CSR module that effectively harnesses and updates the abundant style representation within the source-domain images, and rectifies the target-domain styles to the source domain, thus mitigating the large domain shifts of visual styles under the few-shot setting.
3) To address the challenge of confusing feature alignment arising from sparse objects and complex backgrounds in remote sensing imagery, we propose a CFA

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module. By utilizing fine-grained category-aware masks and performing global feature alignment on the backbone features, this significantly enhances the model's cross-domain detection performance.

We conduct extensive experiments on three cross-domain remote-sensing benchmarks. The results demonstrate that FSDA-DETR achieves superior performance and robust domain-adaptation capability compared with state-of-the-art methods. Specifically, FSDA-DETR achieves a mean average precision (mAP) of 60.5% on the cross-satellite benchmark (xView [24] → DOTA [25]). In the synthetic-to-real benchmark (GTAV10k [26] → UCAS_AOD [27]), FSDA-DETR demonstrates performance comparable to that of supervised learning, reaching an mAP of 81.9%. Furthermore, on the optical-to-synthetic aperture radar (SAR) benchmark (HRRSD [28] → SSDD [29]), the model achieves an mAP of 74.5%, underscoring its marked effectiveness in few-shot cross-domain settings.

II. Related Work

A. Few-Shot Object Detection

Few-shot object detection (FSOD) aims to empower models to rapidly acclimate to new targets by utilizing only a limited quantity of annotated samples. Current FSOD methodologies can be broadly classified into those grounded in meta-learning [30], [31], [32], [33] and those anchored in transfer learning [34], [35]. Meta-learning involves training models across various tasks to promote swift adaptation, with significant contributions, including FSRW [30] and Meta R-CNN [31]. Conversely, transfer learning entails fine-tuning pretrained models with a limited volume of new training data. Exemplary studies in this domain include the TFA [36] and FSCE [34]. Notably, these methodologies are tailored primarily for natural scenes. In contrast, remote sensing imagery, due to its intricate backgrounds and pronounced scale variations, significantly diverges from natural scene images. Consequently, the development of an effective FSOD algorithm that accommodates the unique characteristics of remote sensing scenarios is imperative. To the best of authors' knowledge, the FSODM [37] constitutes an inaugural study focused on FSOD within the realm of remote sensing. In this study, Kang et al. [30] developed a multiscale object detection framework that integrates feature reweighting and bounding box prediction, thereby achieving effective generalization for new classes with only a few samples. Subsequently, Cheng et al. [38] enhanced the recognition of foreground regions through the incorporation of prototype information. ST-FSOD [39] incorporates self-training mechanisms into the few-shot fine-tuning process and employs pseudo-labels to adapt to a wider array of novel object classes. SAM&BFS [40] utilizes a shared attention module to facilitate the model's ability to detect new objects that exhibit considerable size variations. Bou et al. [41] improved interclass distinction learning by fine-tuning class prototypes. TEMO [42] integrates text modality features to mitigate classification ambiguity among novel classes. G-FSDet [43] devises a metric-based discriminative loss to construct a more distinctive classifier. SAE-FSDet [44] establishes a label consistency classifier to rectify the inconsistent allocation of new instances. The InfRS [45] employs a prototype calibration strategy designed to alleviate catastrophic forgetting. Nonetheless, nearly all the previously mentioned methods ideally presuppose that the training and testing datasets originate from the same domain, inadequately addressing the disparities in cross-domain distributions. Unlike these approaches, we present a novel architecture for FSOD in remote sensing imagery, signifying a groundbreaking exploration in this domain.

B. Unsupervised Domain-Adaptive Object Detection

Unsupervised domain-adaptive object detection (UDAOD) seeks to transfer knowledge from a model trained in the source domain to an unlabeled target domain. DAFaster [7] represents the first effort to implement domain-adaptive object detection within the Faster R-CNN framework, aligning domains at both the image and instance levels while incorporating consistency regularization to facilitate the learning of domain-invariant region proposal networks (RPNs) [3]. In parallel, other extensions of Faster R-CNN have been proposed to tackle different challenges. For instance, Zhang et al. [46] proposed a multistage enhancement network to improve tiny object detection in remote sensing images by optimizing instance-level sample assignment and enriching contextual feature representations across multiple detection stages. Inspired by DAFaster, subsequent studies [47], [48] have further optimized feature alignment strategies, thereby continuously enhancing the performance of cross-domain object detection. Recently, approaches employing image-to-image translation [49], self-labeling techniques [50], and graph reasoning [51] have been applied to domain-adaptive object detection tasks. Unlike the aforementioned methods, subsequent approaches utilizing the teacher–student framework [52], [53] generate pseudo-labels for unlabeled target domain images through a teacher model to guide the training of the student model. MRT [54] subsequently endeavors to achieve end-to-end domain-adaptive object detection within the Deformable-DETR [16] architecture, thereby facilitating better extraction of domain-invariant features by the model.

However, research on UDAOD within remote sensing scenarios remains relatively scarce. FADA [55] mitigates the confounding challenge of feature alignment [56]. RFA-Net [57] enhances instance-level feature extraction and alignment processes. In contrast, DualDA-Net [58] further investigates the potential of unlabeled target-domain data within the Faster R-CNN framework. In contrast to the preceding methods, the remote sensing teacher [59] represents the first introduction of a teacher–student model and self-training framework within the transformer architecture. Unfortunately, the aforementioned UDA methods exhibit suboptimal performance when labeled data are scarce in the target domain. Given the limited availability of labeled data in the target domain, we propose an FSDAOD method specifically tailored for remote sensing images.

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C. Few-Shot Domain-Adaptive Object Detection

Under conditions where only a limited number of labeled samples from the target domain and adequately labeled samples from the source domain are available to bridge the domain shifts, FSDAOD integrates the concepts of FSOD and UDAOD to address the challenge of adapting the object detector to the target domain in the presence of scarce labeled data. Notably, the limited labeled samples in the target domain differ in distribution from those in the source domain, yet they share the same label space. To the authors' knowledge, the FAFRCNN [14] is the first study to conduct research on FSDAOD and alleviates the issue of over-adaptation to the source domain caused by an imbalanced data distribution by introducing a feature pairing mechanism. In contrast to FAFRCNN, PICA [10] proposes a novel instance-level alignment method that combines pointwise alignment and moving average centroid alignment. Recently, CutMix [60] has integrated data synthesis methods with adversarial learning. Thus, AcroFOD [11] further introduces adaptive strategies and multilevel domain-aware data augmentation techniques. Like AcroFOD, AsyFOD [12] innovatively proposes an asymmetric adaptation paradigm. As DETR-based detectors demonstrate powerful feature representation capabilities and adaptability to complex scenes in cross-domain object detection tasks, Fu et al. [61] innovatively introduced the CD-ViTO model.

However, the inherent significant domain shifts in remote sensing images (e.g., optical and SAR images) restrict the advancement of these methods. To resolve the aforementioned issues, OS-DAFSD [13] has been tailored within the AcroFOD framework to enable effective knowledge transfer from optical images to SAR images. Nevertheless, this method focuses solely on the ship detection task in SAR images, failing to consider the applications in diverse cross-domain scenarios. Therefore, we present an FSDAOD method derived from DETR-based detectors that effectively overcomes the challenge of significant domain shifts under conditions of few-shot labeled data.

D. Cross-Domain Style Transfer

The cross-domain style transfer [62] module converts the style information of source-domain objects into arbitrarily new styles, ensuring that the detector remains style-invariant and well-generalized, with widespread applications across diverse tasks. In classification tasks, Kang et al. [63] propose an iterative style synthesis methodology, which constructs styles that are distinct from both observed source styles and previously synthesized styles, thereby substantially enhancing the model's domain generalization performance. For segmentation tasks [64], Liu et al. [65] proposed a style selection and style transfer module configured with local binary pattern matching, effectively addressing appearance inconsistencies in ultrasound images. In detection tasks, Rodriguez and Mikolajczyk [66] utilized a style transfer methodology to align the source-domain image pixels with the target domain, thereby ensuring consistent features across varying styles.

In few-shot domain-adaptive tasks, the cross-domain style transfer module facilitates robust generalization to target-domain styles by projecting target-domain style representations into the source-domain's feature space. Fu et al. [67] proposed a methodology that integrates the wavelet transform and style augmentation to enable the exchange of low-frequency cross-domain style information, effectively mitigating domain shifts inherent to cross-domain few-shot learning scenarios. Su et al. [19] introduced a local–global perturbation methodology, which strategically applies perturbations to the local and global statistics of the source domain, facilitating the emulation of distinct target-domain styles and significantly enhancing the model's ability to adapt to target domains.

In this article, we further explore the application of cross-domain style transfer in FSDAOD tasks. Unlike the style augmentation and perturbation methods described above, the CSR module ingeniously utilizes the style discrepancies between the source domain and few-shot target-domain samples to establish a more expressive and discriminative representation space. Subsequently, the module applies momentum updates of stored source-domain style information to rectify the target-domain styles, which significantly enhances the generalization performance to the target domain.

III. Methodology

FSDA-DETR is an effective FSDAOD method based on the DETR framework for remote sensing imagery. In this section, we first articulate the research problem in Section III-A1, revisit the DETR framework in Section III-A2, and introduce the architecture of the detector in Section III-A3. Subsequently, we provide a detailed description of the proposed CSR module and CFA module in Sections III-B and III-C. Finally, we summarize the loss functions employed in this study in Section III-D.

A. Preliminary

1) Problem Setting: We consider a scenario involving a substantial quantity of labeled data from the source domain $D^s = \{(x_i^S, y_i^S)\}_{i=1}^{N^s}$ and a limited amount of labeled data from the target domain $D^t = \{(x_j^t, y_j^t)\}_{j=1}^{N^t}$. Specifically, $x_i, x_j^t \in X$ denote the ith image in the source domain D^s and jth image in the target domain D^t, respectively. $y_i^s, y_j^t \in Y$ represent the associated bounding box coordinates and object class labels for images in the source and target domains, respectively, forming pairs $(C_{\text{source}}, C_{\text{target}})$. FSDAOD aims to train a detector from the abundant source domain D^s that can effectively adapt to the data-scarce target domain D^t (where $N^s \gg N^t = \text{eight}$). Our focus is on scenarios with substantial domain shifts in the mappings $P^s : \mathcal{X} \times Y \to R_s^+$ and $P^t : \mathcal{X} \times Y \to R_t^+$.

To mitigate the performance degradation of the detector caused by domain shifts, our objective is to train a detector $\phi : X \to Y$ that can learn feature representations from the abundant source-domain samples D^s and generalize effectively to the target domain D^t. Unlike conventional FSOD methods based on meta-learning, which typically adopt a "K-shot, N-way" configuration—where "K" denotes the number of labeled instances per class and "N" represents the total number of distinct classes—our method addresses the challenge of high

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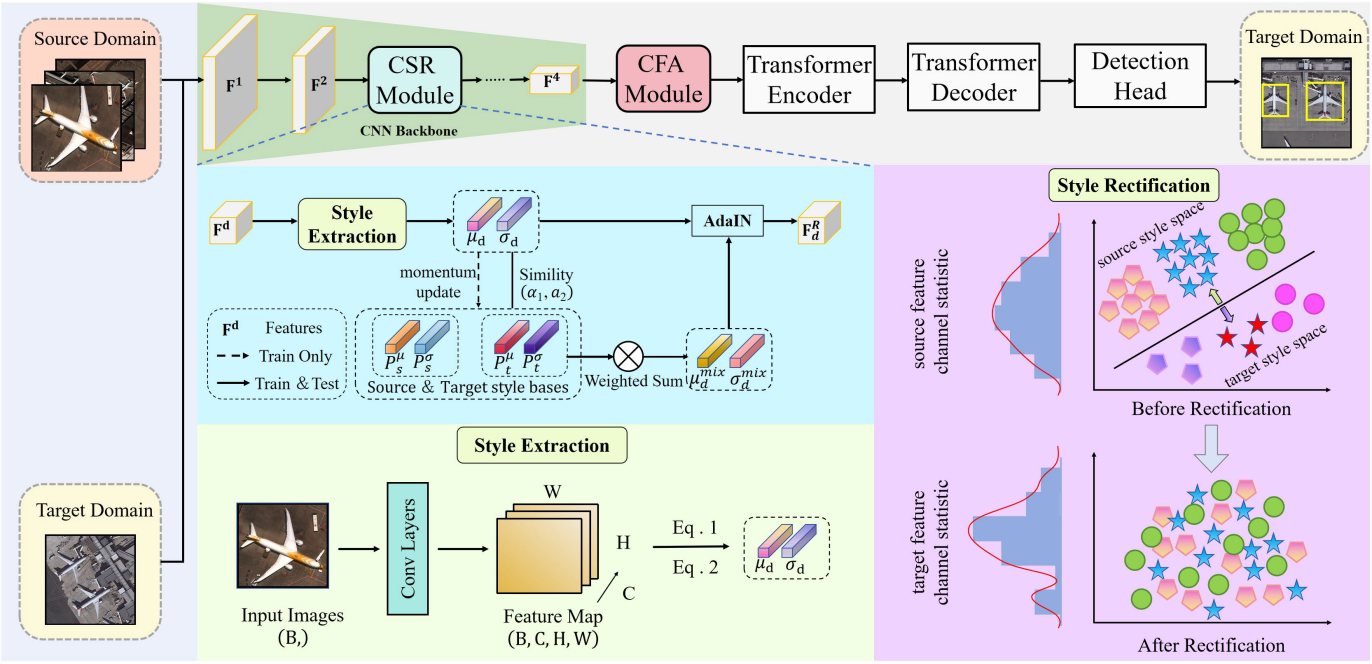


Fig. 2. Details of the FSDA-DETR framework, which leverages DINO as its foundational baseline and incorporates both the CSR module and the CFA module. The CSR module is applied to shallow features within the backbone, facilitating cross-domain style adaptation during training by updating the global style of the source domain through momentum updating. During inference, the trained model directly rectifies the target-domain style.

annotation costs in remote sensing. Therefore, we limit the target domain D^t to only eight labeled and randomly selected training samples. Without loss of generality, we assume that the source domain (with ample training data) and the target domain (with limited labeled data) share the same label space ($C_{\text{source}} = C_{\text{target}}$), but differ in data distribution [i.e., $P^S(x_i^S) \neq P^t(x_j^t)$].

2) DETR Revisit: The DETR introduces an advanced end-to-end framework for object detection that extracts image features through a CNN backbone and feeds the features, with added positional encodings, into the encoder. The decoder embeds object queries as inputs and applies cross-attention to the encoded features, ultimately generating detection predictions through a feed-forward network (FFN). This design eliminates the need for traditional handcrafted components, such as anchor design [68] and NMS, which significantly streamline the object detection pipeline. Building on this foundation, DETR with improved denoising anchor boxes for end-to-end object detection (DINO) further refines the detection process by employing a hybrid query selection method that selectively amplifies location queries with top-K feature associations, thereby preserving their learnability and enabling the encoder to capture a more comprehensive range of content features. Additionally, DINO [17] introduces a denoising training strategy to prevent the model from repeatedly outputting the same target, effectively accelerating convergence and enhancing overall performance.

3) Framework Overview: As shown in Fig. 2, FSDA-DETR employs DINO as the base detector and incorporates CSR and CFA modules. Specifically, to maximize the utilization of information from both source-domain and limited target-domain samples, we pair source- and target-domain images, inputting them into the model for mixed training, thereby improving the cross-domain adaptability of the detector. Each batch contains an equal number of source- and target-domain images. Following feature extraction on each paired image set by the ResNet backbone, we represent the image style through channelwise feature statistics (i.e., mean and variance) and utilize the CSR module to extract style information from the backbone output. Since style information tends to diminish at deeper feature levels, we restrict processing to shallow features. During network training, source-domain image styles are iteratively updated and stored as style representations, dynamically rectifying the target-domain style to source-domain style representations. Subsequently, the CFA module is applied to the rectified source-domain and target-domain feature maps. The CFA module masks specific category feature layers and inputs the masked source- and target-domain feature maps into a domain discriminator D to compute adversarial loss, thereby enhancing cross-domain feature alignment by confusing the discriminator. DINO refines the features further through the encoder, whereas the decoder initiates a mixed-query setup on the refined features from the encoder. Finally, the FFN generates predictions for bounding boxes and object classes based on the decoder's refined features. Importantly, the CFA module is applied exclusively during training. Additional details regarding the CSR and CFA modules are presented in Sections III-B and III-C, respectively.

B. CSR Module

In the FDA setting, where only a limited amount of labeled target images are available, Gao et al. [11] employed an adaptive data augmentation strategy to expand the constrained target-domain data. Although this approach partially mitigates data limitations, it inevitably increases the computational burden of training and may restrict the detector to

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augmented target instances, thereby limiting its generalizability to previously unseen target objects. Prior research has shown that feature distribution shifts, caused by style discrepancies, predominantly occur within the shallow layers of the network [23], [69]. Moreover, shallow feature distributions capture the stylistic information of input images effectively. Consequently, existing methods commonly represent image style distributions through the channelwise mean and variance of shallow features [70]. Building on this insight, we design a CSR module during training, fully utilizing the limited labeled target-domain samples. This module rectifies the target-domain's feature channel statistics (style) to those of the source-domain training set, thereby mitigating the model's maladaptive response to the target domain and facilitating style transfer within the FDA setting.

Specifically, as shown in Fig. 2, the network input consists of paired images from the source domains D^s and D^t. For a given feature map $F_s \in R^{B\times C\times H\times W}$ of the source domain D^s and a feature map $F_t \in R^{B\times C\times H\times W}$ of the target domain D^t (where B, C, H, and W denote the batch size, channel dimension, height, and width of the feature maps, respectively). We calculate the shallow feature channel statistics for both D^s and D^t, specifically the mean and variance for each instance channel in the source and target domains, as follows:

$$\mu_d^c(F_s, F_t) = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W} F_d \tag{1}$$

$$\sigma_d^c(F_s, F_t) = \sqrt{\frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\left(F_d - \mu_d^c(F_s, F_t)\right)^2 + \eta} \tag{2}$$

where $c \in \{C \mid c = 256, 512\}$ represents the number of shallow feature channels in $R^{B\times C\times H\times W}$; $\mu_d^c(F_s, F_t)$ and $\sigma_d^c(F_s, F_t)$ denote the mean and variance statistics calculated from the source domains F_s and F_t, respectively; and η is a smoothing term to prevent zero error.

By applying the above calculation methods, we obtain the mean and variance for each channel c of D^s and D^t. In each training iteration, we continuously optimize the global style statistics from $D^s \rightarrow D^t$ through a momentum update mechanism. Specifically, we have

$$P_{d+1}^{\mu} = \gamma P_d^{\mu} + (1-\gamma)\mu_d^c(F_s, F_t), \quad P_s^{\mu}, P_t^{\mu} \in P_d^{\mu} \tag{3}$$

$$P_{d+1}^{\sigma} = \gamma P_d^{\sigma} + (1-\gamma)\sigma_d^c(F_s, F_t), \quad P_s^{\sigma}, P_t^{\sigma} \in P_d^{\sigma} \tag{4}$$

where $\gamma \in [0, 1]$ denotes the momentum update factor, which is typically assigned a value near 1 to promote stability and continuity in the updating process. The variables P_d^{μ} and P_d^{σ} are initially configured as standard distributions with a mean of 0 and a variance of 1. After the momentum update, P_{d+1}^{μ} and P_{d+1}^{σ} represent the style features of D^s and D^t, respectively.

Serving as the foundation for subsequent style rectification steps, the feature channel statistics of the source and target domains are continuously updated and retained within the model, thereby facilitating its gradual adaptation to cross-domain style variations. Specifically, we designate the source-domain style features P_s^{μ} and P_s^{σ} as style bases, aligning the style features of the input target domain to those of the source-domain training set.

Furthermore, to quantify the distributional divergence between the target- and source-domain features, we introduce the Wasserstein distance [71]—a metric derived from optimal transport theory—used to evaluate the style shift between the style baselines of the target and source domains

$$W_m = \|P_t^{\mu} - P_s^{\mu}\|_2^2 + \left(\left(P_s^{\sigma}\right)^2 + \left(P_t^{\sigma}\right)^2 - 2P_s^{\sigma}\cdot P_t^{\sigma}\right) \tag{5}$$

where W_m denotes the distribution distance between the target- and source-domain style bases. Based on the computed style divergence W_m between the input target-domain features and each source-domain style base, we calculate similarity weights. These weights are subsequently applied to perform a weighted fusion of the source-domain style baselines, thereby rectifying the target-domain features. The specific procedure is outlined as follows:

$$\alpha_m = \frac{\exp(1/1 + W_m)}{\sum_{m=1}^{M}\exp(1/1 + W_m)} \tag{6}$$

where α_m denotes the similarity weight associated with the source-domain style bases and M represents the total count of source-domain style bases. All similarity weights are normalized to form a probability distribution, constrained to sum to 1. Given that target-domain features may be influenced by multiple source-domain style prototypes—particularly in the FDA setting, where target-domain features typically do not correspond precisely to a single source-domain style—we perform a weighted fusion of source-domain style prototypes based on the similarity weights α_m. The process is formalized as follows:

$$\mu_{\mathrm{s}}^{\mathrm{mix}} = \sum_{m=1}^{M}\alpha_m \cdot P_s^{\mu} \tag{7}$$

$$\sigma_{\mathrm{s}}^{\mathrm{mix}} = \sum_{m=1}^{M}\alpha_m \cdot P_s^{\sigma} \tag{8}$$

where μ_s^{mix} and σ_s^{mix} represent the mean and variance of the fused source domain, respectively. Finally, we employ AdaIN [23] to rectify the target-domain style, harmonizing it with the stylistic characteristics of the source domain. The formulation is presented as follows:

$$\mathrm{AdaIN}(F_t) = \frac{F_t - \mu_t^c}{\sigma_t^c}\cdot\sigma_{\mathrm{s}}^{\mathrm{mix}} + \mu_{\mathrm{s}}^{\mathrm{mix}} \tag{9}$$

where μ_t^c, σ_t^c, and F_t denote the mean, variance, and feature map of the target-domain input images, respectively. During both the training and inference phases, the CSR module is employed to refine the style attributes of shallow features within the target-domain images. This module rectifies the stylistic attributes of the target domain to the style bases of the source domain, thereby narrowing the domain shifts of visual styles. Subsequently, the source-domain features and the rectified target-domain features are cohesively transmitted to the subsequent layer of the network, thereby attenuating stylistic disparities between the source and target domains.

C. CFA Module

Within the FDA setting, a singular style rectification strategy proves inadequate for comprehensively addressing

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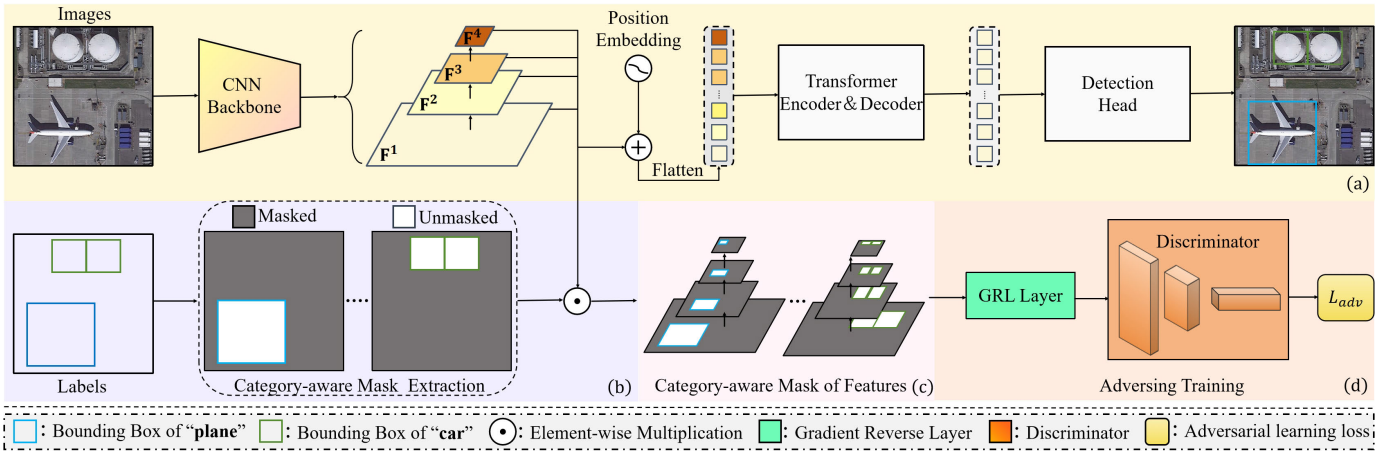


Fig. 3. Details of (a) proposed detection pipeline. Our proposed CFA module encompasses the following components: (b) category-aware mask extraction, (c) category-aware mask features—referring to category-specific mask features derived from the network backbone, and (d) adversarial training. Collectively, these components synergistically promote the alignment of foreground features within different categories, thereby contributing to a significant augmentation of domain adaptation capability.

domain shift challenges in cross-domain object detection. Furthermore, existing feature alignment methods are prone to interference from confounding attributes in remote sensing images, such as the sparsity of targets and the complexity of backgrounds, which consequently impede the precise alignment of instance-level foreground features across domains. In response, FSDA-DETR integrates a CFA module, which is meticulously engineered to facilitate instance-level foreground alignment with category specificity, as detailed in the following. The module structure is shown in Fig. 3

1) Category-Aware Mask Extraction: In the framework of object detection tasks, with annotations available in both the source domain and the few-shot target domain, we utilize bounding boxes and class labels to generate precise, category-aware masks, which serve to enhance the cross-domain adaptability of foreground instances at the instance level. Specifically, given a set of images (D^s and D^t), each target object's class label is represented by the set $L = \{C_1, C_2, \ldots, C_N\}$, where C_i denotes the class of the ith object. For each specified category C_i, we define a corresponding set of bounding boxes, $B_{C_i} = \{b_1^{C_i}, b_2^{C_i}, \ldots, b_n^{C_i}\}$, with each bounding box $b_i^c = (x_{\min}^i, y_{\min}^i, x_{\max}^i, y_{\max}^i)$ specifying the coordinates of the region of interest, from the top-left to the bottom-right corners. Furthermore, we construct $Y = \{(C_1, B_{C_1}), (C_2, B_{C_2}), \ldots, (C_N, B_{C_N})\}$ to encapsulate the comprehensive annotation information for all target objects across the image set, thus enabling a unified framework for subsequent mask generation and classification. Leveraging B_{C_i}, we obtain the full set of bounding boxes associated with C_i, upon which a binary classification mask for each category is then established as follows:

$$M^c(x, y) = \begin{cases} 1, & \text{if } (x, y) \in R_i; \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

where $R_i = \{(x, y) \mid x_{\min}^i \le x \le x_{\max}^i, y_{\min}^i \le y \le y_{\max}^i\}$ represents the spatial domain of category c and $M^c(x, y)$ denotes the binary mask for this category. Here, $M^c(x, y) = 1$ signifies the designated region in the image corresponding to the responsive area for category c, ensuring spatial alignment for subsequent feature processing and analysis.

2) Category Mask Matching for Hierarchical Features: To achieve precise alignment of instance-level foreground features, the extracted category mask $M^c(x, y)$ is initially applied to the multilevel feature representations within both the source and target domains. This process preserves only the pertinent foreground regions corresponding to each category. Specifically, the category mask is downsampled to conform to the spatial dimensions of each feature level l_i, as expressed by the following equation:

$$M_c'(l_i) = \phi\left(M^c(x, y), H_{l_i}, W_{l_i}\right) \tag{11}$$

where H_{l_i} and W_{l_i} denote the height and width of the l_ith feature level, respectively, and "ϕ" represents the downsampling function. Subsequently, the downsampled category mask $\mathbf{M}_c'(l_i)$ is applied in an elementwise multiplication to the corresponding feature map at each level, as defined by the following equation:

$$F_{l_i}' = F_{l_i} \circ M_c'(l_i) \tag{12}$$

where "$\circ$" indicates the elementwise multiplication and F_{l_i} denotes the original feature map of the l_ith level. Through this process, the category mask $M_c'(l_i)$ is incorporated into the feature maps F_l^S and F_l^T of both domains D^s and D^t, yielding the masked feature maps $(F_{s_l}', F_{t_l}' \in F_{l_i}')$.

3) Category-Aware Feature Alignment: To further strengthen the detector's cross-domain detection performance on the few-shot target domain, we perform category-aware, fine-grained feature alignment on the feature maps F_{s_l}' and F_{t_l}' following the extraction of category-specific masks. Specifically, category masks for D^s and D^t are applied at multiple feature levels, which are subsequently fed into the domain discriminator D, which is structured as a multilayer CNN. The discriminator discerns whether each feature originates from the source domain (denoted by $d = 0$) or the target domain (denoted by $d = 1$) . Furthermore, to enhance the model's robustness to domain variations, we incorporate

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a gradient reversal layer (GRL) [72] during adversarial training to promote domain invariance. Finally, we propose a binary cross-entropy loss function, which is designed based on category awareness, to optimize the adversarial learning process. The loss function is formulated as follows:

$$L_{\text{source}} = -\sum_{l=1}^{L}\sum_{c_{l_i}\in\mathcal{C}} d\log\left(D\left(F'_{s_l}\right)\right) \tag{13}$$

$$L_{\text{target}} = -\sum_{l=1}^{L}\sum_{c_{l_i}\in\mathcal{C}} (1-d)\log\left(1-D\left(F'_{t_l}\right)\right) \tag{14}$$

$$L_{\text{adv}} = L_{\text{source}} + L_{\text{target}} \tag{15}$$

where $D(F'_{s_l})$ and $D(F'_{t_l})$ denote the outputs of the discriminator corresponding to the feature map F'_{l_i}, L denotes the total number of feature levels within the backbone, and l refers to the index of each feature level (with $l = 1, 2, \ldots, L$). The set C denotes the collection of categories present in each batch, while c_{l_i} specifies an individual category within C.

D. Network Training

To guarantee that the proposed CSR and CFA modules substantively bolster the detector's domain adaptation proficiency, FSDA-DETR integrates two synergistic loss functions throughout the training process. First, a supervised detection loss L_{det}, which is consistent with DINO, is integrated to ensure model stability and reliability across cross-domain detection tasks. Subsequently, an adversarial loss L_{adv} is incorporated within the CFA module to facilitate the alignment of backbone foreground features, effectively mitigating feature distribution discrepancies between the source and target domains. The total loss function is formulated as follows by combining these loss functions:

$$L = L_{\text{det}} + \lambda_c L_{\text{adv}} \tag{16}$$

where λ_c denotes the balancing coefficient for the adversarial loss, modulating its relative contribution to the supervised loss during the optimization of the overall detection objective. The training workflow of our method is illustrated in Algorithm 1.

Algorithm 1 Training Pipeline of the FSDA-DETR

Input: Initialized detector ϕ^{ini}, labeled source dataset $D^s = \{(x_i^s, y_i^s)\}_{i=1}^{N^s}$, few labeled target dataset $D^t = \{(x_j^t, y_j^t)\}_{j=1}^{N^t}$, max epochs E, iterations for every epoch K, hyperparameter λ_c, adversarial loss L_{adv}, supervised detection loss L_{det}, total loss L.

Output: Domain Adaptive Detector $f_\phi = (g_\phi, h_\phi)$ consists of the backbone g_ϕ and detect head h_ϕ

```
Initialize φ ← φ^ini
Initialize backbone g_φ
Initialize discriminator D
for epoch = 1 to E do
  for iteration = 1 to K do
    Sample batch B^s = {(x_i^s, y_i^s)}_{i=1}^{N^s} ∈ D^s
    Sample batch B^t = {(x_j^t, y_j^t)}_{j=1}^{N^t} ∈ D^t
    Get feature (F_s, F_t) ∈ (B_s, B_t) with g_φ
    Extract style {μ_d(F_s, F_t), σ_d(F_s, F_t)} ∈ {D^s, D^t}
    using Eq. (1)-(2)
    Store and update source domain style via Eq. (3)-(8)
    Rectify the target domain style using Eq. (9)
    Compute L_adv via D and Eq. (13)-(15)
    Predictions pred = f_φ(x_i^s, x_j^t)
    Compute detection loss L_det = (pred, {y_i^s, ..., y_j^t})
    Compute total losss L = L_det + λ_c L_adv in Eq. (16)
    Backpropagate and update φ to minimize L
  end for
end for
```

IV. Experiments

A comprehensive overview of our experiments is presented in this section, including datasets and scenarios, evaluation metrics, implementation details, ablation studies, comparisons with state of the art, performance on different data magnitudes, visualization and analysis, and conclusion. A detailed discussion of each part is provided in the following subsections.

A. Datasets and Scenarios

In this article, we rigorously curate six representative cross-domain remote sensing datasets to construct and simulate three commonly encountered domain adaptation benchmarks. These datasets are leveraged to systematically evaluate the domain adaptability and generalization capacity of FSDA-DETR under diverse cross-domain scenarios. First, following the configurations of studies [59] and [73], we evaluate the model's performance in cross-domain knowledge transfer within the satellite imagery benchmark (xView [24] → DOTA [25]). Subsequently, adhering to the settings in [26], we validate the proposed method's efficacy on synthetic-to-real benchmarks (GTAV10k [26] → UCAS_AOD [27]). Lastly, in alignment with studies [73] and [74], we explore the model's cross-domain-adaptive capability between the optical and SAR domains using SAR benchmarks (HRRSD [28] → SSDD [29]). Detailed descriptions of these datasets are provided in the following sections.

1) DOTA: The DOTA dataset, a comprehensive large-scale dataset for optical remote sensing imagery, was released by Wuhan University in 2018. This dataset encompasses 2806 high-resolution aerial images obtained from sources such as Google Earth, each with a resolution of approximately 4000 × 4000 pixels. The DOTA dataset comprises 188 282 annotated instances spanning 15 distinct object classes, including categories such as aircraft, vessels, and vehicles. These images exhibit extensive variations in scale, orientation, and morphology, with annotations in the form of arbitrary quadrilaterals to capture the diversity and complexity inherent in aerial scenes more effectively. Consequently, the DOTA dataset is employed as the target domain in our experiments to evaluate the cross-domain generalization capabilities of FSDA-DETR. To maintain coherence with the categorical structure of the xView dataset, which is designated as the source domain, this study focuses on three primary object categories: plane, storage tank, and ship. The model is trained on the officially

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defined training set, and its performance is evaluated on the validation set. The original images are cropped into 800 × 800 patches, with a 100-pixel overlapping stride.

2) xView: Unlike the DOTA dataset, captured via Google Earth satellites, the xView dataset is derived from the WorldView-3 satellite and comprises images with resolutions mainly between 2500 × 2500 and 5000 × 3000 pixels, collectively covering over 1400 km^2 of Earth's surface. This extensive dataset includes 60 distinct object categories and approximately one million meticulously annotated instances, distributed across 1129 high-resolution satellite images with a spatial resolution reaching 0.3 m. xView encompasses substantial geographic and categorical diversity, presenting scenes from dense urban environments to remote rural regions. Following the methodologies outlined in [59] and [73], we designate the xView dataset as the target domain and pair it with the DOTA dataset to rigorously evaluate cross-domain adaptability and knowledge transfer efficacy across distinct satellite sources. Our experiments are conducted using the official training set, focusing specifically on three primary categories: plane, storage tank, and ship. The original images are cropped into 800 × 800 patches, with a 100-pixel overlapping stride.

3) GTAV10k: The GTAV10k dataset, a synthetic remote sensing dataset tailored for vehicle detection in cross-domain adaptation tasks, encompasses 10 001 images annotated with 56 193 vehicle bounding boxes. Each image, rendered at a resolution of approximately 1280 × 1024 pixels, depicts a wide variety of scenes and vehicle distributions. In contrast to real-world images, the GTAV10k dataset comprises game-engine-generated imagery with an imaging perspective highly consistent with the conventional nadir view in aerial photography, while providing superior image resolution. The game environment's extensive virtual maps and diverse imaging conditions result in a comprehensive synthetic image distribution. For this study, GTAV10k serves as the source-domain dataset. Following the cross-domain configuration defined in [26], it is paired with the UCAS_AOD dataset, facilitating a rigorous assessment of the model's cross-domain adaptability from synthetic to real-world benchmark data.

4) UCAS_AOD: The UCAS_AOD dataset, developed by the University of Chinese Academy of Sciences (UCAS) and first released in 2014 with additional updates in 2015, comprises high-resolution images sourced from Google Earth. The dataset contains meticulously selected images designed to ensure a uniform distribution of target objects across diverse orientations. Serving as a synthetic-to-real benchmark, UCAS_AOD encompasses two categories: automobiles and aircraft. This study specifically focuses on the automobile category. The dataset comprises 510 images with resolutions of 1280 × 659 and 1372 × 941 pixels, totaling 7114 annotated instances of vehicles. Given the appropriate resolution, no further image preprocessing through cropping was deemed necessary for this study. Moreover, as the UCAS_AOD dataset lacks a standardized official split, we established a custom configuration for the training and testing subsets. To ensure the reproducibility of the experimental results, images with filenames ending in the digits 3, 6, and 9 were allocated to the test set, while the remaining images were designated for training, preserving a 7:3 ratio between the training and test sets.

5) HRRSD: The HRRSD dataset, released by Chinese Academy of Sciences in 2019, aims to advance research on high-resolution remote sensing imagery for object detection. This dataset comprises 21 761 images sourced from Google Earth and Baidu Maps, with resolutions ranging from 0.15 to 1.2 m, encompassing diverse geographical environments and complex scenes. The dataset contains 55 740 annotated object instances across 13 typical remote sensing categories. In this study, we focus specifically on the subset of ship images within this dataset, which includes 2163 images with a total of 3971 annotated ships, all of which are utilized exclusively for training. Furthermore, for the cross-domain adaptation task between optical and SAR images, the HRRSD dataset is designated as the source domain in our cross-domain adaptation experiments.

6) SSDD: The SSDD dataset, designed specifically for ship detection in SAR imagery, was made available by the Naval Aeronautical and Astronautical University of China. It comprises 1160 images and 2456 annotated ship instances, with an average of 2.12 ships per image, covering a variety of complex scenes ranging from open seas to coastal regions, under varying weather conditions, sea states, and ship types. Following the approach in [73] and [74], we establish SSDD as the target domain, pairing it with the HRRSD dataset to evaluate the cross-domain adaptation effectiveness between optical and SAR imagery. Our experiments are conducted on the officially designated training set, with evaluation performed on the corresponding test set.

B. Evaluation Metrics

In this article, we define the average precision (AP) at a intersection over union (IoU) threshold of 0.5 as the class-specific AP50 and the mAP as the mAP50. Additionally, the AP or mAP represents the mean of the precision values at ten different IoU thresholds (ranging from 0.5 to 0.95, with an increment of 0.05), providing a comprehensive measure of the model's detection performance across various IoU thresholds. Specifically, the AP is a commonly used metric for evaluating object detection performance, as it integrates the AP across different IoU thresholds (0.5–0.95) and the corresponding recall levels, serving as a balanced assessment of the algorithm's performance at different overlap levels. It is mathematically expressed as follows:

$$\mathrm{AP} = \int_0^1 P(R)\mathrm{d}R. \tag{17}$$

The mAP aggregates the precision–recall curves across different object categories, calculates their average values, and thus provides a comprehensive evaluation of the algorithm's overall performance on the entire dataset. It is mathematically expressed as follows:

$$\mathrm{mAP} = \frac{1}{c}\sum_{i=0}^{c} \mathrm{AP}_i. \tag{18}$$

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TABLE I
ABLATION STUDY OF FSDA-DETR: xView → DOTA

Source-only	Target	CSR	CFA	Plane	Storage-tank	Ship	mAP50(%)
✓				64.6	35.4	58.0	52.7
	✓			94.8	78.2	74.2	82.4
✓	✓			65.0	39.6	60.5	55.0
✓	✓	✓		66.1	45.1	56.3	55.8
✓	✓		✓	67.2	39.7	**62.8**	56.6
✓	✓	✓	✓	**69.2**	**50.4**	61.8	**60.5**

C. Implementation Details

In our experiment, we implement our methodology on the DINO detection framework and utilize ResNet-50 pretrained on ImageNet [77] as the backbone network. Furthermore, we compare our methodology against existing frameworks of UDAOD and FSDAOD. To ensure the fairness of the comparison, we employ the entire dataset of the target domain to report the respective results within the UDA setting. First, in the FDA setting, the eight fully annotated images from the target domain employed for training were consistently maintained across all experimental iterations (detailed information of the eight randomly selected images is provided in the code). Additionally, under the "N-way K-shot" setting, we set the number of targets in each image according to different shot numbers ($K = 3$, 5, and 10), following [13]. For all the experiments, we standardize the batch size to 2 for both the source and target domains. Our model achieves substantial performance after a training duration of only 36 training epochs. For three cross-domain scenarios, the balancing coefficient λ_c in (14) is set to 0.1. Consistent with the DINO setup, we employ the Adam optimizer [78] to train our model, initializing the learning rate 2×10^{-4} and applying a decay factor of 0.1 at the 30th epoch. All the experimental procedures are executed on an NVIDIA Tesla V100 GPU with 32 GB of memory.

D. Ablation Studies

To investigate the contribution of each component, we perform an ablation study by incrementally adding the CSR module and CFA module to the baseline. Following [11] and [12], the experiments are conducted using the fully annotated target-domain training set (eight randomly selected images from DOTA) and all source-domain training set (xView), with subsequent testing on the official validation set of DOTA. More specifically, among the eight randomly selected and fully annotated target-domain images, there are 140 box-level annotations, including 25 plane annotations, 23 storage-tank annotations, and 92 ship annotations. Furthermore, each image is an 800 × 800 slice extracted from the original target-domain images, and to ensure the reproducibility of our results, we have recorded the exact positions of these slices in our code.

1) Effectiveness of Each Component: The corresponding results are illustrated in Table I. The "Source-only" uses only source-domain data for training, whereas the "Target" includes target-domain data. Training DINO on source-domain data alone poses challenges in achieving optimal performance, primarily due to domain shifts and the scarcity of labeled target data, as illustrated in row 1 of Table I. It can be seen that the superior performance achieved by training on the officially partitioned target training sets is primarily attributed to the DINO's robust generalization ability. The following can be illustrated in Table I.

1) The result listed in row 3 indicates that utilizing target-domain data in training facilitates the detector's adaptation to the target domain, achieving 55.0% mAP with a 2.3% improvement.
2) As can be seen in row 4, our proposed CSR demonstrates satisfactory performance improvement in the cross-satellite adaptation scenarios. Due to the CSR module, the detector can suitably benefit from the rectified target-domain style, achieving a 3.1% mAP gain in FDA tasks. By mitigating significant domain shifts in visual styles, the CSR module fully demonstrates its effectiveness in adapting to the target domain.
3) In comparison to CSR, our proposed CFA module utilizes adversarial learning to align the output features of the backbone, achieving a significant 3.9% improvement, which demonstrates that it encourages detector to focus on foreground object regions, thereby enhancing cross-domain detection performance.
4) The incremental usage of each component enhances cross-domain detection performance, resulting in a 7.8% improvement, as shown in row 6. This demonstrates that the CSR module mitigates domain shifts at the image style level and that the CFA module promotes the mitigation of foreground domain shifts in the CNN backbone.

TABLE II
ABLATION STUDY OF THE HYPERPARAMETER λ_c

Method	λ_c	mAP_{50}
DINO (ICLR'23) [17]	—	52.7
DINO + CFA	0.01	53.4
	0.05	55.4
	0.1	**56.6**

TABLE III
COMPARISON OF COMPUTATIONAL COMPLEXITY BETWEEN VANILLA DINO AND FSDA-DETR

Method	Image size	Param (MB)	FLOPs (G)	Time (ms)
DINO (ICLR'23) [17]	800^2	45.1	**177.8**	**210.3**
FSDA-DETR (Ours)	800^2	**46.7**	178.6	212.8

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2) Hyperparameter Sensitivity Analysis: To balance the adversarial learning loss and detection loss, the hyperparameter λ_c serves as the weighting coefficient in the proposed CFA module, as defined in (16). In the cross-satellite benchmark (xView [24] → DOTA [25]), we set λ_c to 0.01, 0.05, and 0.1 to evaluate its impact on our CFA module, as described in [53]. The variation in overall accuracy is illustrated in Table II. Notably, our proposed CFA module effectively helps the detector in improving the cross-domain detection performance, thereby contributing to domain alignment under various weight settings. Furthermore, it can be concluded from the table that increasing the weights leads to a slight performance improvement, due to the significant contribution of the CFA module during training. In this article, we set the weight to 0.1 to align with established methodologies [7], [75], thereby improving the interpretability of our CFA module and ensuring a fair comparison.

3) Analysis of the Computational Complexity: During the training process, our proposed CSR and CFA as additional plugins to tackle the task of FDA. To confirm that our approach does not incur additional computational overhead, we make a further comparison of computational complexity between Vanilla DINO and FSDA-DETR on the cross-satellite benchmark (xView [24] → DOTA [25]). Table III presents the statistical results, including model parameters, FLOPs, and inference time, where our method adds only a minimal parameters budget. In addition, the CFA module is employed solely during training and removed at inference, thereby exhibiting the same execution speed and computational efficiency as the vanilla DINO. This experiment indicates that our method strikes an effective balance between model complexity and performance.

4) Exploring the Impact of Different Network Hierarchies on Style Rectification: As shown in Table IV, we further analyze the role of backbone feature distributions at different levels in style transfer performance. The experimental results indicate that performance progressively deteriorates as network depth increases. This trend can be attributed to the gradual dispersion of stylistic information in deeper network layers, which diminishes the effectiveness of style transfer.

TABLE IV
ABLATION STUDY OF DIFFERENT NETWORK HIERARCHIES

Source-only	Target	CSR	Different Layers	mAP50(%)
✓			-	52.7
	✓		-	82.4
✓	✓	✓	layer1	55.3
✓	✓	✓	layer2	**55.8**
✓	✓	✓	layer3	54.6
✓	✓	✓	layer4	53.4

E. Comparisons With State of the Arts

To evaluate the domain adaptation capability of FSDA-DETR under data scarcity, we assess its performance across three representative cross-domain adaptation benchmarks that we construct: 1) cross-satellite benchmark, in which we perform knowledge transfer from xView to DOTA, thereby assessing the model's generalization ability across distinct satellite images by utilizing the xView dataset for training the model and subsequently evaluating its performance on DOTA; 2) synthetic-to-real benchmark, where we train the model on GTAV10k and evaluate it on the UCAS_AOD target domain; and 3) for the optical-to-SAR benchmark, in which we train the model on HRRSD, followed by testing it on SSDD.

1) Cross-Satellite Benchmark: Table V lists the three categories' detection accuracy of the proposed method and other state-of-the-art algorithms on the target-domain set. DINO outperforms CNN-based detectors in cross-domain scenarios. Our method significantly outperforms other methods, with "ship" category results far exceeding those of existing FSDAOD methods, proving the CSR and CFA modules can effectively strengthen the detector's cross-domain detection performance under the few-shot setting. Meanwhile, our method attains the mAP of 60.5%, exceeding that of the previous state-of-the-art UDAOD method, AQT, by 2.2%. Though cross-satellite scenarios are more challenging due to sparse objects and complex background noise, our CSR and CFA modules significantly mitigate the domain shifts by aligning the target domain with the source domain.

2) Synthetic-to-Real Benchmark: Table VI summarizes the results of various methods on UCAS_AOD datasets. It can be intuitively concluded from the performance improvements under the synthetic-to-real benchmark that our CSR module effectively mitigates domain shifts under the significant visual discrepancies between synthetic and real images. Moreover, the proposed FSDA-DETR achieves 81.9% in terms of mAP50, which is 6.1% higher than the current state-of-the-art method AsyFOD under the same setting with minimal labeled target data and approaches the performance of supervised learning methods in synthetic-real scene.

3) Optical-to-SAR Benchmark: Table VII shows the detection performance on the SSDD dataset by UDAOD and FSDAOD methods. The experimental results indicate that our proposed method outperforms the state-of-the-art methods under a more challenging setting of optical-to-SAR domain adaptation, increasing the mAP from 27.5% to 74.5%, a boost of 7.8% compared with AsyFOD. This performance improvement can be attributed to CFA's ability that is general to different cross-domain adaptions, which is further augmented by CSR module.

In conclusion, comprehensive experimental outcomes across three distinct cross-domain benchmarks indicate that our FSDA-DETR demonstrates remarkable generalizability capabilities under both minor and large domain shifts. Furthermore, it effectively mitigates domain shifts, even under conditions of extreme scarcity in target-domain data, underscoring the robustness and efficacy of the proposed methodology.

F. Performance Under the Few-Shot Setting

Following [13], we further present the comparative experimental results of FSDA-DETR under the few-shot setting across three cross-domain scenarios to verify its generalization

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TABLE V
EXPERIMENTAL RESULTS (%) OF THE SCENARIO BETWEEN DIFFERENT SATELLITES: xView → DOTA

xView → DOTA									
Setting	Method	Type	Backbone	Source	Target	Plane	Storage-tank	Ship	mAP50(%)
-	DINO (ICLR'23) [17]	Transformer	ResNet-50	✓		64.6	35.4	58.0	52.7
		Transformer	ResNet-50		✓	94.8	78.2	74.2	82.4
UDAOD	DAFaster (CVPR'18) [7]	CNN	VGG16	✓		57.3	36.4	34.3	42.7
	AQT (IJCAI'22) [75]	Transformer	ResNet-50	✓		68.2	46.8	59.9	58.3
	FADA (TGRS'22) [55]	CNN	ResNet-50	✓		68.0	46.2	43.4	52.5
	CMT (CVPR'23) [76]	CNN	ResNet-101	✓		66.7	56.4	38.5	53.8
	MRT (ICCV'23) [54]	Transformer	ResNet-50	✓		67.3	39.9	55.7	54.3
FSDAOD	FAFRCNN (CVPR'19) [14]	CNN	VGG16	✓		46.9	31.5	32.8	37.1
	PICA (WACV'22) [10]	CNN	ResNet-50	✓		48.4	39.1	38.0	41.8
	AsyFOD (CVPR'23) [12]	CNN	YOLOv5-x	✓		61.8	**66.7**	23.4	50.6
	OS-DAFSD (TGRS'24) [13]	CNN	YOLOv5-n	✓		66.4	65.8	35.2	55.8
	FSDA-DETR (Ours)	Transformer	ResNet-50	✓		**69.2**	50.4	**61.8**	**60.5**

TABLE VI
EXPERIMENTAL RESULTS (%) OF THE SCENARIO BETWEEN SYNTHETIC AND REAL: GTAV10K → UCAS_AOD

GTAV10k → UCAS_AOD						
Setting	Method	Type	Backbone	Source	Target	AP50(%)
-	DINO (ICLR'23) [17]	Transformer	ResNet-50	✓		67.3
		Transformer	ResNet-50		✓	88.8
UDAOD	DAFaster (CVPR'18) [7]	CNN	VGG16	✓		59.0
	AQT (IJCAI'22) [75]	Transformer	ResNet-50	✓		58.6
	FADA (TGRS'22) [55]	CNN	ResNet-50	✓		57.8
	CMT (CVPR'23) [76]	CNN	ResNet-101	✓		56.2
	MRT (ICCV'23) [54]	Transformer	ResNet-50	✓		65.9
FSDAOD	FAFRCNN (CVPR'19) [14]	CNN	VGG16	✓		53.7
	PICA (WACV'22) [10]	CNN	ResNet-50	✓		56.2
	AsyFOD (CVPR'23) [12]	CNN	YOLOv5-x	✓		75.8
	OS-DAFSD (TGRS'24) [13]	CNN	YOLOv5-n	✓		70.5
	FSDA-DETR (Ours)	Transformer	ResNet-50	✓		**81.9**

TABLE VII
EXPERIMENTAL RESULTS (%) OF THE SCENARIO BETWEEN OPTICAL AND SAR: HRRSD → SSDD

HRRSD → SSDD						
Setting	Method	Type	Backbone	Source	Target	AP50(%)
-	DINO (ICLR'23) [17]	Transformer	ResNet-50	✓		27.5
		Transformer	ResNet-50		✓	94.9
UDAOD	DAFaster (CVPR'18) [7]	CNN	VGG16	✓		51.9
	AQT (IJCAI'22) [75]	Transformer	ResNet-50	✓		39.9
	FADA (TGRS'22) [55]	CNN	ResNet-50	✓		49.4
	CMT (CVPR'23) [76]	CNN	ResNet-101	✓		53.7
	MRT (ICCV'23) [54]	Transformer	ResNet-50	✓		59.0
FSDAOD	FAFRCNN (CVPR'19) [14]	CNN	VGG16	✓		34.2
	PICA (WACV'22) [10]	CNN	ResNet-50	✓		56.6
	AsyFOD (CVPR'23) [12]	CNN	YOLOv5-x	✓		63.7
	OS-DAFSD (TGRS'24) [13]	CNN	YOLOv5-n	✓		56.7
	FSDA-DETR (Ours)	Transformer	ResNet-50	✓		**74.5**

ability. As shown in Table VIII, FSDA-DETR achieves superior performance compared to existing FSDAOD methods on the cross-satellite benchmark, yielding an mAP50 of 58.5%, 63.6%, and 67.8% in the common three-, five-, and ten-shot settings, respectively. Table IX presents the synthetic-to-real benchmark results, where FSDA-DETR achieves the highest mAP of 77.7%, 80.3%, and 84.1% under three-, five-, and ten-shot settings, respectively. Meanwhile, to further demonstrate the robustness and generalization ability of FSDA-DETR, we present the experimental results under the few-shot setting for the most challenging optical-to-SAR scenario, as presented in Table X. Notably, the experimental results for FSDA-DETR demonstrate a significant performance improvement from three shots to ten shots, with mAP increasing from 73.4% to 83.9%, attaining a 10.5% improvement. These improvements primarily result from the joint contributions of our CSR and CFA modules. The experimental results further demonstrate FSDA-DETR's suitability for addressing FSDAOD tasks in remote sensing imagery.

G. Performance on Different Data Magnitudes

To evaluate the performance of FSDA-DETR across varying amounts of available target-domain data, we conducted experiments on the challenging optical-to-SAR benchmark. Specifically, utilizing the officially designated SSDD dataset,

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TABLE VIII

COMPARISON OF DOMAIN-ADAPTIVE DETECTION METHODS IN CROSS-SATELLITE BENCHMARK UNDER K-SHOT SETTING

xView → DOTA												
Setting	Method	3-shot				5-shot				10-shot			
		Plane	Storage-tank	Ship	mAP50(%)	Plane	Storage-tank	Ship	mAP50(%)	Plane	Storage-tank	Ship	mAP50(%)
FSDAOD	FAFRCNN (CVPR'19) [14]	40.7	45.7	29.4	38.6	42.8	37.7	33.7	38.1	45.8	45.1	43.6	44.8
	PICA (WACV'22) [10]	47.5	40.9	40.5	42.9	55.4	55.8	44.2	51.5	61.4	53.0	51.8	55.4
	AsyFOD (CVPR'23) [12]	59.8	41.3	**59.0**	53.4	61.1	40.8	**76.7**	59.5	66.2	45.9	**68.6**	60.2
	OS-DAFSD (TGRS'24) [13]	59.1	50.0	54.9	54.6	61.8	61.1	58.5	60.5	67.3	58.4	61.8	62.5
	FSDA-DETR (Ours)	**64.9**	**52.1**	58.4	**58.5**	**66.3**	**61.8**	62.6	**63.6**	**73.9**	**63.5**	66.2	**67.8**

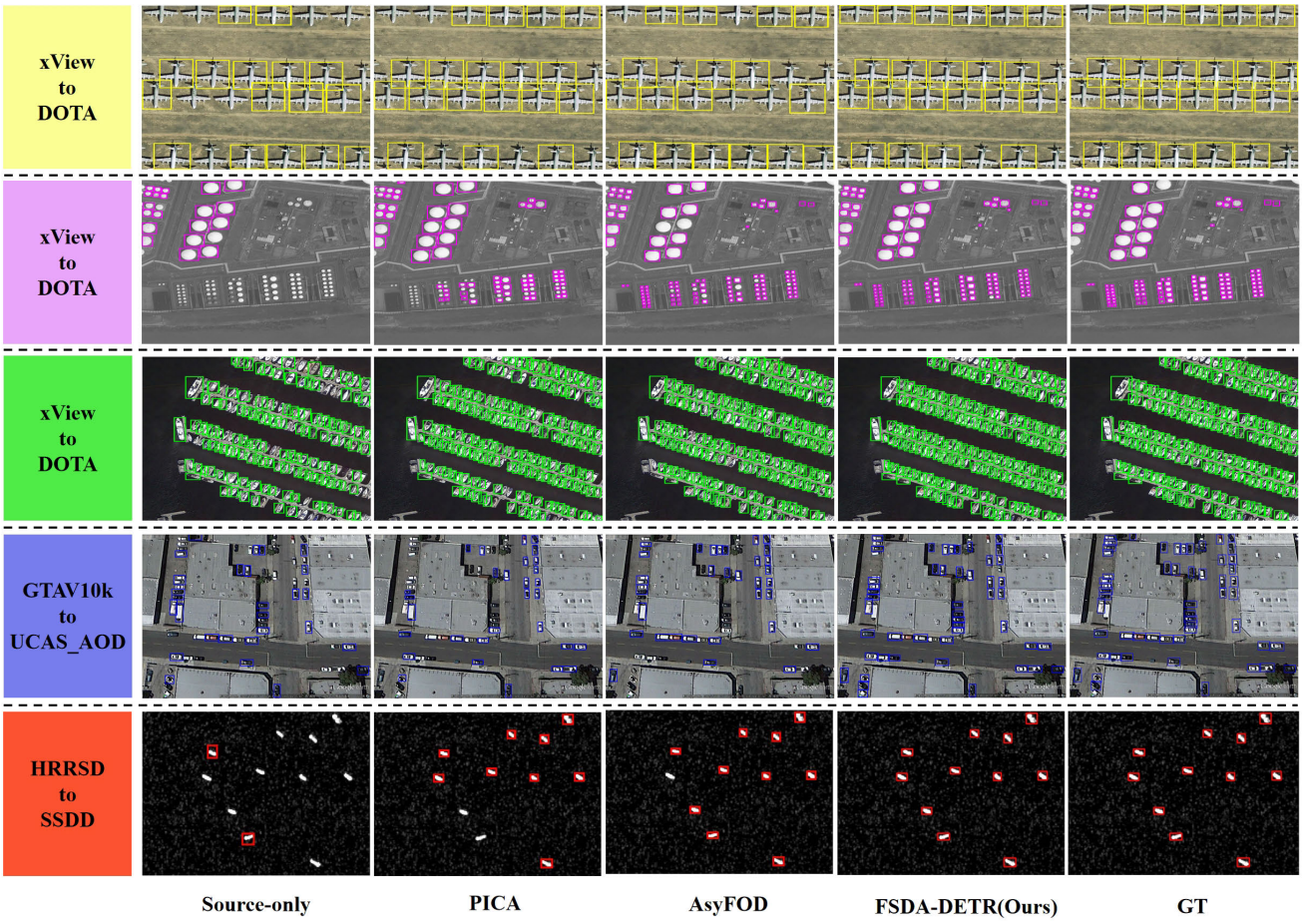


Fig. 4. Visualization results of the comparison between our method and state-of-the-art methods under three cross-domain benchmarks. Different categories are represented by distinct colors. The confidence threshold set at 0.2. "Source-only" denotes the results from training on source-domain data only. "GT" indicates the ground-truth annotations.

TABLE IX

COMPARISON OF DOMAIN-ADAPTIVE DETECTION METHODS IN SYNTHETIC-TO-REAL TRANSFER UNDER K-SHOT SETTING

GTAV10k → UCAS_AOD				
Setting	Method	Car mAP50(%)		
		3-shot	5-shot	10-shot
FSDAOD	FAFRCNN (CVPR'19) [14]	47.1	50.6	57.8
	PICA (WACV'22) [10]	51.8	52.1	60.2
	AsyFOD (CVPR'23) [12]	69.6	72.9	78.3
	OS-DAFSD (TGRS'24) [13]	68.1	71.6	72.6
	FSDA-DETR (Ours)	**77.7**	**80.3**	**84.1**

TABLE X

COMPARISON OF DOMAIN-ADAPTIVE DETECTION METHODS IN CROSS-MODALITY OPTICAL–SAR UNDER K-SHOT SETTING

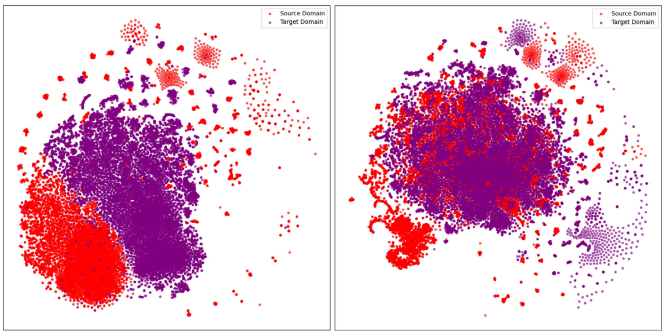
HRRSD → SSDD				
Setting	Method	Ship mAP50(%)		
		3-shot	5-shot	10-shot
FSDAOD	FAFRCNN (CVPR'19) [14]	33.5	36.9	53.2
	PICA (WACV'22) [10]	54.6	58.0	62.1
	AsyFOD (CVPR'23) [12]	63.5	66.2	71.6
	OS-DAFSD (TGRS'24) [13]	57.9	62.7	69.5
	FSDA-DETR (Ours)	**73.4**	**75.1**	**83.9**

we randomly selected 0.2%, 2%, and 20% of the training set, which comprises 928 images, for model training and these selections correspond to 2, 18, and 185 images, respectively. As shown in Table XI, FSDA-DETR consistently demonstrates superior performance compared to AsyFOD [12] across all the experimental scenarios, with performance gains becom-

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TABLE XI
RESULTS (%) OF MORE DATA IN THE SCENARIO BETWEEN OPTICAL AND SAR

HRRSD → SSDD					
Proportion	Method	Source	Target	AP(%)	AP50(%)
-	DINO (ICLR'23) [17]	✓		7.5	22.5
			✓	70.1	94.9
0.2%	DINO (ICLR'23) [17]		✓	-	-
	AsyFOD (CVPR'23) [12]	✓		23.4	47.2
	FSDA-DETR (Ours)	✓		**28.4**	**53.5**
2%	DINO (ICLR'23) [17]		✓	-	-
	AsyFOD (CVPR'23) [12]	✓		41.6	68.6
	FSDA-DETR (Ours)	✓		**52.5**	**84.5**
20%	DINO (ICLR'23) [17]		✓	50.4	77.3
	AsyFOD (CVPR'23) [12]	✓		51.8	82.5
	FSDA-DETR (Ours)	✓		**60.1**	**90.3**



(a) (b)

Fig. 5. t-SNE visualizations of shallow backbone layers under the cross-satellite benchmark, before and after style rectifications, are shown, with red for the source domain and purple for the target domain. Compared to vanilla DINO, the CSR module effectively alleviates stylistic appearance shifts. (a) DINO. (b) FSDA-DETR.

ing increasingly pronounced as the amount of training data increases, which also proves the robustness of our proposed method.

H. Visualization and Analysis

1) Detection Results: The detection result of source-only [17], PICA [10], AsyFOD [12], FSDA-DETR (ours), and GT across three cross-domain benchmarks are shown in Fig. 4. In cross-domain scenarios with large domain shifts, our method exhibits superior performance over other FSDAOD methods. Notably, the results in rows 2 (xView → DOTA) and 5 (HRRSD → SSDD) indicate that our method achieves performance approximating the ground truth, exhibiting superior classification accuracy and higher recall rates. This finding indicates that FADA-DETR effectively mitigates domain shifts across diverse data-scarce adaptation scenarios.

2) Feature Visualization: Using the t-distributed stochastic neighbor embedding (t-SNE) [79] method, we analyze the effectiveness of the proposed CSR module. Fig. 5 illustrates the feature distributions of the shallow backbone layers under a cross-satellite benchmark, both before and after the application of the CSR module. The vanilla DINO distinctly separates domain-specific styles. Following style rectification, the CSR module effectively aligns the styles of the source and target domains, significantly alleviating stylistic appearance shifts.

V. CONCLUSION

In this article, we introduce FSDA-DETR, an end-to-end few-shot domain-adaptive framework based on DINO. To the best of authors' knowledge, this is the first study to explore the application of DETR-style detectors for FSDAOD within remote sensing. Initially, we introduce a CSR module that stores and dynamically updates the weighted-fusion source-domain style bases and rectifies the target-domain styles accordingly, effectively mitigating the domain shifts of visual styles between the source and target domains. Furthermore, we introduce a CFA module, which performs fine-grained masking on object regions of rectified backbone features corresponding to different categories and utilizes adversarial training to extract domain-invariant features, which further enhances the detector's cross-domain detection performance. The introduction of these two modules significantly enhances the model's generalization capability of our method to the target domain in few-shot domain-adaptive settings. The comprehensive results obtained from three established cross-domain adaptation benchmarks illustrate that FSDA-DETR achieves significant performance gains in FSDAOD tasks, reaching state-of-the-art performance. In the future, we would like to extend FSDA-DETR to adapt to scenarios such as building extraction [80] and change detection [81] in remote sensing imagery, where data annotation is costly and domain shifts persist across varying geographic contexts and sensor conditions.

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